P11.5 Impurity site at $n=0$

$$H_{n,m} = \varepsilon_0 \delta_{n,m} - t(\delta_{n+1,m} + \delta_{n-1,m}) + \Delta \delta_{n,0}\delta_{n,m}$$

(a) $\phi_n = A e^{-qa|n|}$ exponentially localized solution.

Consider $n \geq 2$ (so that impurity site $n=0$ is not involved):

$$(E - \varepsilon_0)\phi_n = -t(\phi_{n-1} + \phi_{n+1}) \quad . \text{ For } n \geq 2,\ \underset{n-1}{|n-1|} = \cancel{1},\ \underset{n+1}{|n+1|} = \cancel{1}$$

$$\Rightarrow \quad E - \varepsilon_0 = -t(e^{-qa} + e^{+qa}) \qquad (1)$$

Now $n=0$: $(E - \varepsilon_0 - \Delta)\phi_0 = -t(\phi_1 + \phi_{-1})$

$$E - \varepsilon_0 - \Delta = -t\, 2 e^{-qa} \qquad (2)$$

Use (1): $-t(e^{-qa} + e^{+qa}) - \Delta = -2t\, e^{-qa}$

$$\Delta = t(e^{-qa} - e^{qa}) = -2t \sinh(qa) \Rightarrow \underline{\Delta < 0}$$

for the localized solution.

Let $e^{qa} = x \Rightarrow \quad x - \frac{1}{x} + \frac{\Delta}{t} = 0$

$$x^2 + \frac{\Delta}{t}x - 1 = 0$$

$x = \frac{1}{2}\left(-\frac{\Delta}{t} \pm \sqrt{\left(\frac{\Delta}{t}\right)^2 + 4}\right)$ and, since $e^{qa} > 0$, we must choose (+) sign.

$$e^{qa} = -\frac{\Delta}{2t} + \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2}$$

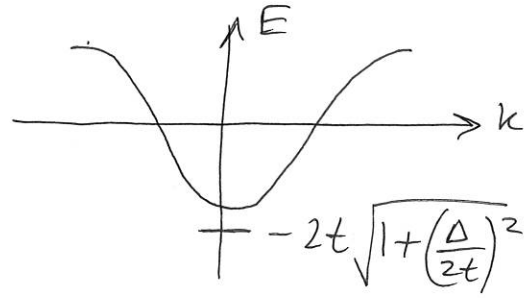
The energy of this state is

$$E = \varepsilon_0 - t(e^{qa} + e^{-qa}) = \varepsilon_0 - t\left\{-\frac{\Delta}{2t} + \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2} + \right.$$

$$\left. + \frac{1}{-\frac{\Delta}{2t} + \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2}}\right\}$$

Use $\dfrac{1}{-\frac{\Delta}{2t} + \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2}} = \dfrac{\frac{\Delta}{2t} + \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2}}{-\left(\frac{\Delta}{2t}\right)^2 + 1 + \left(\frac{\Delta}{2t}\right)^2} = \sqrt{1 + \left(\frac{\Delta}{2t}\right)^2} + \frac{\Delta}{2t}$

Thus $E = \varepsilon_0 - 2t\sqrt{1+\left(\frac{\Delta}{2t}\right)^2}$ lies below the minimum of the conduction band for which $E(k) = \varepsilon_0 - 2t\cos ka$.



$$-2t\sqrt{1+\left(\frac{\Delta}{2t}\right)^2} \approx -2t - \frac{\Delta^2}{4t} \quad \text{for } |\Delta| \ll 2t.$$

(b) $$H\psi = -\frac{\hbar^2}{2m^*}\frac{d^2\psi}{dx^2} + a\Delta\,\delta(x)\,\psi(x) = E\psi$$

As usual, integrate both sides over $\int_{-\delta}^{\delta} dx$ and take $\delta \to 0$ limit:

$$-\frac{\hbar^2}{2m^*}\left(\frac{d\psi}{dx}\Big|_{\delta} - \frac{d\psi}{dx}\Big|_{-\delta}\right) + a\Delta\psi(0) = E\int_{-\delta}^{\delta} dx\,\psi(x) \xrightarrow[\delta\to 0]{} 0$$

$$\psi'(+0) - \psi'(-0) = \frac{2m^* a\Delta}{\hbar^2}\psi(0)$$

Let $\psi = A e^{-\varkappa|x|} \Rightarrow \psi'(+0) = A\frac{d}{dx}\left(e^{-\varkappa x}\right)_{x=0} = -\varkappa A$

$$\psi'(-0) = A\frac{d}{dx}\left(e^{\varkappa x}\right)_{x=0} = +\varkappa A$$

$$\Rightarrow -2\varkappa A = \frac{2m^* a\Delta}{\hbar^2}A \Rightarrow \varkappa = -\frac{m^* a\Delta}{\hbar^2}$$

$\Rightarrow$ bound solution exists for $\Delta < 0$.

Find energy: away from $x=0$ point $E_b\psi = -\frac{\hbar^2}{2m^*}\frac{d^2}{dx^2}\left(Ae^{-\varkappa x}\right)$ (for $x>0$)

$$\Rightarrow E_b = -\frac{\hbar^2}{2m^*}\varkappa^2 = -\frac{\hbar^2}{2m}\left(\frac{m^* a\Delta}{\hbar^2}\right)^2 = -\frac{m^* a^2\Delta^2}{2\hbar^2}.$$

Effective mass for part (a):

$$E(k) = -2t\cos(ka) \approx -2t\left(1 - \tfrac{1}{2}(ka)^2\right)$$
$$= -2t + ta^2k^2 = -2t + \frac{\hbar^2k^2}{2m^*}$$

$$\Rightarrow \frac{1}{m^*} = \frac{2ta^2}{\hbar^2}, \quad m^* = \frac{\hbar^2}{2ta^2}$$

$$\Rightarrow E_b = -\frac{m^* a^2\Delta^2}{2\hbar^2} = -\frac{\Delta^2}{4t}$$ just as in part (a).

P 11.6 $H = \varepsilon_0 \delta_{n,m} - t(\delta_{n+1,m} + \delta_{n-1,m}) + \Delta \delta_{n,m} \delta_{n,0}$

$$\phi_n = \begin{cases} T e^{-ikna}, & n \geq 0 \\ e^{-ikna} + R e^{ikna}, & n < 0 \end{cases}$$

$$E \phi_{-1} = \varepsilon_0 \phi_{-1} - t(\phi_{-2} + \phi_0) \qquad (1)$$

$$E \phi_0 = (\varepsilon_0 + \Delta)\phi_0 - t(\phi_{-1} + \phi_1) \qquad (2)$$

(1) $$(E - \varepsilon_0)(e^{ika} + R e^{-ika}) = -t(e^{+ik2a} + R e^{-ik2a} + T)$$

But E is function of k, for example consider eqn. for ϕ_{+2}:

$$E \phi_{+2} = \varepsilon_0 \phi_{+2} - t(\phi_{+3} + \phi_{+1})$$

$$\Rightarrow \underline{E - \varepsilon_0 = -2t \cos ka}$$

$\Rightarrow$ (1) gives $$(e^{ika} + e^{-ika})(e^{ika} + R e^{-ika}) = e^{+ik2a} + R e^{-ika} + T$$

$$e^{i2ka} + 1 + R(1 + e^{-i2ka}) = e^{i2ka} + R e^{-i2ka} + T$$

$$\Rightarrow 1 + R = T$$

(2) $$(E - \varepsilon_0 - \Delta) T = -t(e^{ika} + \underbrace{R}_{T-1} e^{-ika} + T e^{-ika})$$

$$= -t(2i \sin ka + T \cdot 2 e^{-ika})$$

$$\Rightarrow (-2t \cos ka + 2t e^{-ika} - \Delta) T = -t \, 2i \sin ka$$

$$-2t\left[\tfrac{1}{2}(e^{ika} + e^{-ika}) - e^{-ika}\right] = -2t \, \tfrac{1}{2} \underbrace{(e^{ika} - e^{-ika})}_{2i \sin ka}$$

$$\Rightarrow (-2t i \sin ka - \Delta) T = -2it \sin ka$$

$$T = \frac{2it\sin ka}{\Delta + 2it\sin ka} = \frac{1}{1 + \frac{\Delta}{i2t\sin ka}}$$

$$T = \frac{1}{1 - \frac{i\Delta}{2t\sin ka}} \Rightarrow |T|^2 = \frac{1}{1 + \left(\frac{\Delta}{2t\sin ka}\right)^2}$$

$$R = T - 1 = \frac{1}{1 - \frac{i\Delta}{2t\sin ka}} - 1 = \frac{\frac{i\Delta}{2t\sin ka}}{1 - \frac{i\Delta}{2t\sin ka}}$$

$$|R|^2 = \frac{\left(\frac{\Delta}{2t\sin ka}\right)^2}{1 + \left(\frac{\Delta}{2t\sin ka}\right)^2}$$

$\Rightarrow |R|^2 + |T^2| = 1 \rightarrow$ current is conserved.

+Some additional scribings, just for fun:

$$T = \frac{1 + \frac{i\Delta}{2t\sin ka}}{1 + \left(\frac{\Delta}{2t\sin ka}\right)^2} \Rightarrow T = 0 \text{ when } \sin(ka) = 0 \Rightarrow k = 0, \pi/a.$$

Find Max(T) as function of k:

$$\frac{\partial T}{\partial k} = \frac{-1}{\left(1 - \frac{i\Delta}{2t\sin ka}\right)^2}\left(-\frac{i\Delta}{2t}\right)\underbrace{\frac{\partial}{\partial k}\frac{1}{\sin ka}}_{\frac{-1}{\sin^2 ka}\cdot a\cos ka} = 0$$

$\Rightarrow \cos ka = 0 \Rightarrow k = \pm\frac{\pi}{2a}$. Group velocity is greatest at that k.

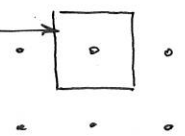
$$\text{Max}(T) = \frac{1}{1 - \frac{i\Delta}{2t}} = \frac{1 + i\,\Delta/2t}{1 + \left(\frac{\Delta}{2t}\right)^2}$$

P15.4 $V(x,y) = 2V_{10}\left(\cos\frac{2\pi x}{a} + \cos\frac{2\pi y}{a}\right) + 4V_{11}\cos\frac{2\pi x}{a}\cos\frac{2\pi y}{a}$

We need to find $V_G = \langle k' | V | k\rangle$ for $\vec{k}' = \vec{k} + G$

$$V_G = \frac{1}{v}\int_{\text{unit cell}} d\vec{x}\; e^{-i\vec{G}\cdot\vec{x}}\, V(\vec{x}) = \frac{1}{a^2}\int_{-a/2}^{a/2} dx \int_{-a/2}^{a/2} dy\; e^{-iG_x x - iG_y y}\, V(x,y)$$

Square lattice



$$\vec{G} = \frac{2\pi}{a} n_1 \hat{x} + \frac{2\pi}{a} n_2 \hat{y}$$

$$\int_{-a/2}^{a/2} dx\; e^{-iG_x x}\cos\frac{2\pi x}{a} = \frac{1}{2}\int_{-a/2}^{a/2} dx\; e^{i\left(\frac{2\pi}{a} - G_x\right)x} + e^{-i\left(\frac{2\pi}{a} + G_x\right)x}$$

$$= \frac{1}{2}\int_{-a/2}^{a/2} dx\; e^{i\frac{2\pi}{a}(1-n_1)x} + e^{-i\frac{2\pi}{a}(1+n_1)x} = \left(\frac{2\pi}{a}x = y\right)$$

$$= \frac{1}{2}\,\frac{a}{2\pi}\int_{-\pi}^{\pi} dy\left(e^{i(1-n_1)y} + e^{-i(1+n_1)y}\right) =$$

$$= \frac{a}{2\pi}\int_0^{\pi} dy\left\{\cos(1-n_1)y + \cos(1+n_1)y\right\} = \frac{a}{2\pi}\,\pi\left(\delta_{n_1,1} + \delta_{n_1,-1}\right).$$

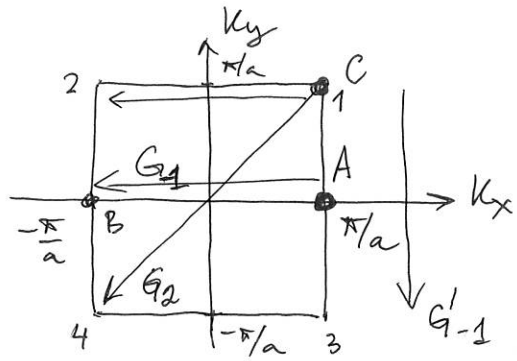
$$\Rightarrow V_G = \frac{1}{a^2}\left\{2V_{10}\,\frac{a^2}{2}\left(\delta_{n_1,1} + \delta_{n_1,-1} + \delta_{n_2,1} + \delta_{n_2,-1}\right)\right.$$

$$\left. + 4V_{11}\left(\frac{a}{2}\right)^2\left(\delta_{n_1,1} + \delta_{n_1,-1}\right)\left(\delta_{n_2,1} + \delta_{n_2,-1}\right)\right\}$$

$$V_G = \begin{cases} V_{10} & \text{for } \vec{G} = \frac{2\pi}{a}(\pm 1, 0) \text{ and } \frac{2\pi}{a}(0, \pm 1) \\ V_{11} & \text{for } \vec{G} = \frac{2\pi}{a}(\pm 1, \pm 1) \end{cases}$$

these signs are not correlated

(a) 1BZ for the square lattice



Point $A = \left(\frac{\pi}{a}, 0\right)$ is connected to $B = \left(-\frac{\pi}{a}, 0\right)$ by $\vec{G}_{-1} = \frac{2\pi}{a}(-1, 0)$.

All other $\vec{G}$'s take it outside the 1BZ.

$$\Rightarrow |\psi\rangle = \alpha\, |\vec{k}_A\rangle + \beta\, |\vec{k}_B\rangle$$

$$|\vec{k}_A\rangle = |\vec{k} = \left(\tfrac{\pi}{a}, 0\right)\rangle$$

$$|\vec{k}_B\rangle = |\vec{k} = \left(-\tfrac{\pi}{a}, 0\right)\rangle = |\vec{k}_A + \vec{G}_{-1}\rangle$$

$\Rightarrow$ compare with Eq. 15.7:

$$\begin{pmatrix} \varepsilon_A & V_{10}^* \\ V_{10} & \varepsilon_B \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\varepsilon_A = \frac{\hbar^2}{2m}\left(\frac{\pi}{a}\right)^2$$

$$\varepsilon_B = \frac{\hbar^2}{2m}\left(-\frac{\pi}{a}\right)^2 = \varepsilon_A = \varepsilon$$

$$\Rightarrow \det \begin{pmatrix} \varepsilon - E & V_{10}^* \\ V_{10} & \varepsilon - E \end{pmatrix} = (\varepsilon - E)^2 - |V_{10}|^2 = 0$$

$$E = \varepsilon \pm |V_{10}|.$$

(b) Point $C = \left(\frac{\pi}{a}, \frac{\pi}{a}\right)$ is connected to 3 other corners of the 1BZ by vectors $\vec{G}_1$, $\vec{G}'_{-1}$ and $\vec{G}_2$.

$\Rightarrow$ we have 4 states:

$$|\psi\rangle = \alpha\, |1\rangle + \beta\, |2\rangle + \gamma\, |3\rangle + \delta\, |4\rangle$$

$$\vec{k} = \left(\tfrac{\pi}{a}, \tfrac{\pi}{a}\right) \qquad \vec{k} = \left(-\tfrac{\pi}{a}, \tfrac{\pi}{a}\right) \qquad \vec{k} = \left(\tfrac{\pi}{a}, -\tfrac{\pi}{a}\right) \qquad \vec{k} = \left(-\tfrac{\pi}{a}, -\tfrac{\pi}{a}\right)$$

$$\varepsilon_1 = \varepsilon_2 = \varepsilon_3 = \varepsilon_4 = \frac{\hbar^2}{2m}\left(\left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{a}\right)^2\right) = \frac{\hbar^2}{m}\left(\frac{\pi}{a}\right)^2 = \varepsilon$$

$$\hat{H} = \begin{pmatrix} \varepsilon & V_{10} & V_{10} & V_{11} \\ V_{10} & \varepsilon & V_{11} & V_{10} \\ V_{10} & V_{11} & \varepsilon & V_{10} \\ V_{11} & V_{10} & V_{10} & \varepsilon \end{pmatrix}$$

$$\det\left(\hat{H} - E\,\hat{I}\right) = 0 \Rightarrow \left(V_{11} - (\varepsilon - E)\right)^2 \left[\left(V_{11} + \varepsilon - E\right)^2 - 4\,V_{10}^2\right] = 0$$

$$\Rightarrow \quad E = \varepsilon - V_{11} \quad \text{(two solutions)}$$

$$\left(E - \varepsilon - V_{11}\right)^2 = 4\,V_{10}^2$$

$$\Rightarrow \quad E = \varepsilon + V_{11} \pm 2\,V_{10}\,.$$